Quiz 4 - Results

Attempt 1 of 1

Written Oct 13, 2026 10:48 AM - Oct 13, 2026 10:53 AM

Attempt Score	**8 / 10 - 80 %**
Overall Grade (Highest Attempt)	**8 / 10 - 80 %**

Question 1 — 2 / 2 points

Take nums = [10, 9, 2, 5, 3, 7, 101, 18]. Which of these is a subsequence but NOT a substring?

- ○ [2, 5, 3]
- ○ [9, 10]
- ○ [5, 2, 7]
- ✓ ◉ [2, 5, 7]

Question 2 — 2 / 2 points

dp[i] = the length of the longest increasing subsequence ending at index i. For nums = [3, 1, 4, 2, 5], what is dp[4] (the 5)?

- ○ 1
- ✓ ◉ 3
- ○ 2
- ○ 5

Question 3 2 / 2 points

Why is the LIS answer max{dp[0], ..., dp[n−1]} and not simply dp[n−1]?

- ○ Because the table is filled from right to left.
- ○ Because dp[n−1] has not been computed when the loop ends.
- ✓ ◉ Because dp[i] is the best subsequence ending at index i, and the best subsequence does not have to end at the last element.
- ○ Because dp[n−1] is always 1.

Question 4 0 / 2 points

What is the edit distance between CAT and CART?

- ✗ ◉ 2
- ○ 3
- ➡ ○ 1
- ○ 0

Question 5 2 / 2 points

In an alignment of two strings, which of these can NOT be the last column?

- ○ a letter over a gap
- ✓ ◉ a gap over a gap
- ○ a letter over a letter
- ○ a gap over a letter

Done